

EXERCISES AND EXACT SOLUTIONS

Maintain First, Replace With Evidence

Independent practice, facilitated discussion, and assessment reference

Production Unit	PU-B07-C56
Notebook	V7_D_N06
Version	0.1
Assessment boundary	Learning evidence—not operational authorization

Instructions

Complete Part A before consulting the solutions. State assumptions explicitly. Where evidence is insufficient, write an abstention or escalation response rather than inventing precision.

Exercise 1

Add network dependencies.

- Required output: a concise decision-ready response with owner, evidence, uncertainty, and boundary where applicable.

Exercise 2

Compare preventive and reactive costs.

- Required output: a concise decision-ready response with owner, evidence, uncertainty, and boundary where applicable.

Exercise 3

Explain the safety boundary.

- Required output: a concise decision-ready response with owner, evidence, uncertainty, and boundary where applicable.

Exact solutions and scoring guide

Solution 1

Model upstream/downstream consequences and single points of failure.

Solution 2

Use lifecycle cost, failure probability, service disruption, and uncertainty.

Solution 3

Only qualified inspectors and responsible authorities can certify, restrict use, procure, or authorize works.

Scoring rubric

Criterion	Strong evidence	Revision required
Decision framing	Owner, action, horizon, authority named	Model output substituted for decision
Evidence	Concepts, grain, provenance, limits explicit	Data treated as self-explanatory
Uncertainty	Abstention/escalation used appropriately	False precision or hidden limitations
Governance	Human authority and safeguards retained	Authority silently delegated